MSG No. 58.402 P_Bnnm [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `..2/m`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,5,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,6,7]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,12,13,16]`
4	$\left[0,\ \frac{1}{2},\ 0\right]$	`[10,11,14,15]`

Table 2: Wyckoff site: `4b`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4,5,8]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,3,6,7]`
3	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[9,12,13,16]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[10,11,14,15]`

Table 3: Wyckoff site: `4c`, site symmetry: `2'm'm`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ 0\right]$	`[1,8,10,15]`
2	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[2,7,9,16]`
3	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[3,6,12,13]`
4	$\left[-x,\ \frac{3}{4},\ 0\right]$	`[4,5,11,14]`

Table 4: Wyckoff site: `8d`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{4},\ 0,\ \frac{1}{4}\right]$	`[1,13]`
2	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,14]`
3	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,15]`
4	$\left[\frac{3}{4},\ 0,\ \frac{1}{4}\right]$	`[4,16]`
5	$\left[\frac{3}{4},\ 0,\ \frac{3}{4}\right]$	`[5,9]`
6	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[6,10]`
7	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,11]`
8	$\left[\frac{1}{4},\ 0,\ \frac{3}{4}\right]$	`[8,12]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,4]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,3]`
3	$[0,\ 0,\ -z]$	`[5,8]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,7]`

continued ...

Table 5

No.	position	mapping
5	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[9,12]
6	$\left[0, \frac{1}{2}, -z\right]$	[10,11]
7	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[13,16]
8	$\left[0, \frac{1}{2}, z\right]$	[14,15]

Table 6: Wyckoff site: 8f, site symmetry: ..m

No.	position	mapping
1	$[x, y, 0]$	[1,8]
2	$\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	[2,7]
3	$\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}\right]$	[3,6]
4	$[-x, -y, 0]$	[4,5]
5	$\left[x+\frac{1}{2}, y, \frac{1}{2}\right]$	[9,16]
6	$\left[x, \frac{1}{2}-y, 0\right]$	[10,15]
7	$\left[-x, y+\frac{1}{2}, 0\right]$	[11,14]
8	$\left[\frac{1}{2}-x, -y, \frac{1}{2}\right]$	[12,13]

Table 7: Wyckoff site: 8g, site symmetry: .m'.

No.	position	mapping
1	$\left[x, \frac{1}{4}, z\right]$	[1,15]
2	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	[2,16]
3	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{2}-z\right]$	[3,13]
4	$\left[-x, \frac{3}{4}, z\right]$	[4,14]
5	$\left[-x, \frac{3}{4}, -z\right]$	[5,11]
6	$\left[\frac{1}{2}-x, \frac{3}{4}, z+\frac{1}{2}\right]$	[6,12]
7	$\left[x+\frac{1}{2}, \frac{1}{4}, z+\frac{1}{2}\right]$	[7,9]
8	$\left[x, \frac{1}{4}, -z\right]$	[8,10]

Table 8: Wyckoff site: 16h, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[x+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[2]
3	$\left[\frac{1}{2}-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[3]
4	$[-x, -y, z]$	[4]
5	$[-x, -y, -z]$	[5]
6	$\left[\frac{1}{2}-x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[6]
7	$\left[x+\frac{1}{2}, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[7]
8	$[x, y, -z]$	[8]
9	$\left[x+\frac{1}{2}, y, z+\frac{1}{2}\right]$	[9]
10	$\left[x, \frac{1}{2}-y, -z\right]$	[10]
11	$\left[-x, y+\frac{1}{2}, -z\right]$	[11]

continued ...

Table 8

No.	position	mapping
12	$\left[\frac{1}{2}-x,\,-y,\,z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x,\,-y,\,\frac{1}{2}-z\right]$	[13]
14	$\left[-x,\,y+\frac{1}{2},\,z\right]$	[14]
15	$\left[x,\,\frac{1}{2}-y,\,z\right]$	[15]
16	$\left[x+\frac{1}{2},\,y,\,\frac{1}{2}-z\right]$	[16]